

Design Document

Project 1 - Personal Homepage

CS 5610 Web Development, Northeastern University

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Project Description

For this project I built my personal homepage. The idea was to make a page that shows who I am as a backend and infrastructure engineer, the projects I have built, and the kind of roles I am looking for. It is a front-end only static site, written in plain HTML5, CSS3, and JavaScript (ES6 modules). I did not use any front-end framework, jQuery, or component library. The layout uses flexbox and CSS grid, and the whole thing is deployed on GitHub Pages.

I gave the site an infrastructure and terminal theme because it matches what I actually work on. The home page opens with an animated terminal that types out a short intro like a shell session, which is my original component for this project. The site has three pages, each at its own URL:

- Home (index.html): the animated terminal intro, a short tagline, my tech stack, and links.
- About (about.html): my background, work experience, featured projects, and a contact form with client-side validation.
- Projects (projects.html): a gallery of my projects. Each project has an architecture diagram, and you can filter by technology, sort the list, and open a project to see more detail. This is the page I built with the help of AI.

User Personas

I thought about three main people who would actually land on this page.

Priya - Technical Recruiter

Priya looks at a lot of candidates every day for backend and platform roles, and she only spends about a minute on each one. She wants to see my name, my degree, and my core tech stack right away without digging for it. She often checks profiles on her phone between meetings, so the page has to work well on mobile and load fast. If she cannot tell what I do in the first few seconds, she moves on.

David - Senior Engineer / Hiring Manager

David gets the candidates Priya passes along. He does not care about flashy design. He wants to click into my actual GitHub repos and see whether my projects are real systems or just toy apps. He cares about clear writing and honest scope, and he will open a few projects in new tabs to read the code and the READMEs.

Sam - Fellow Student / Peer

Sam is a classmate in the same program. He finds my site from a shared link or the class showcase. He is curious how I built the terminal animation and the projects gallery, and he might want to connect on LinkedIn or look at my source code. He appreciates a clean front end that is easy to inspect.

User Stories

- As a recruiter, I want to see the candidate's name, program, and main tech stack the moment the page loads, so I can decide quickly if they fit a role.
- As a recruiter on my phone, I want the page to reflow cleanly on a small screen, so I can read it between meetings.
- As a hiring manager, I want each project to link straight to its GitHub repo, so I can check that the work is real.
- As a hiring manager, I want a short description and the tech stack for each project, so I can judge the depth without opening every repo.
- As a peer, I want to filter the projects by technology, so I can find the ones that use something I care about.
- As a peer, I want to open a project and see its architecture, so I can understand how it is put together.
- As a visitor, I want an intro that stands out (the terminal), so the page feels memorable instead of templated.
- As a visitor using a screen reader, I want alt text on images and proper semantic HTML, so the site is accessible to me.

Design and Mockups

I started from rough wireframes for the three pages before writing any code, sketching out where the nav, the terminal, the content, and the footer would go. The final pages grew from those sketches and picked up more detail as I built, especially the projects gallery. The screenshots below show the design as it turned out.

Home page

A dark nav bar, then the terminal that types out whoami, my bio, and my project list, followed by my tagline, tech badges, and two buttons. This is the first thing a visitor sees.

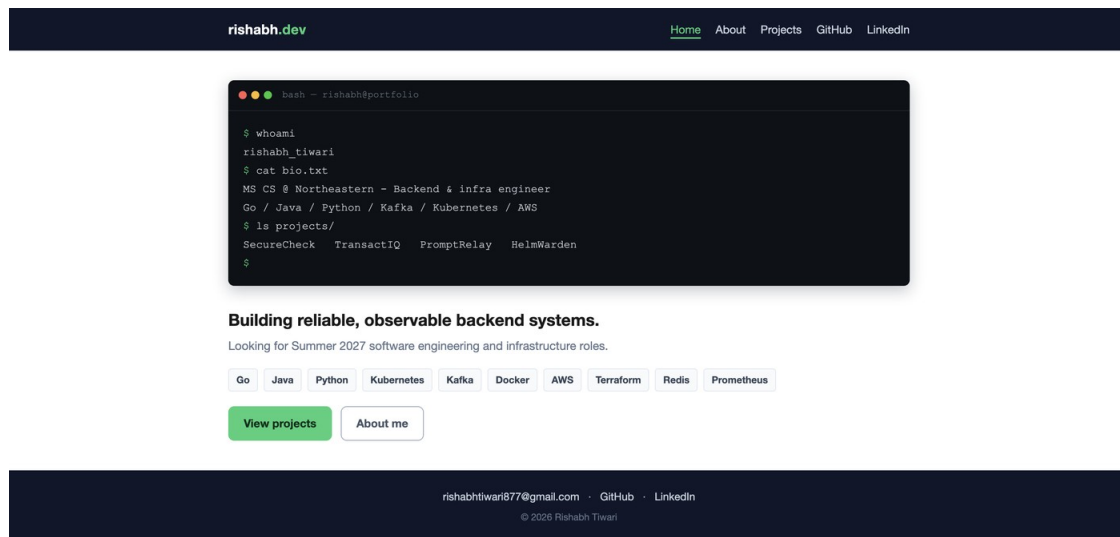


Figure 1. Home page with the animated terminal.

About page

My photo and a short bio at the top, then an experience timeline, featured project cards, and a contact form. This is the page that goes into more depth about my background.



Rishabh Tiwari

Backend & Infrastructure Engineer

I am a Master of Science in Computer Science student at Northeastern University's Khoury College, graduating December 2027. I focus on backend and infrastructure engineering: distributed systems, event-driven pipelines, and the tooling that keeps them observable and reliable. I currently work as a Teaching Assistant for DS 3000, and I have shipped production features across two software engineering internships.

Experience

Teaching Assistant — DS 3000 Khoury College, Northeastern University · Sep 2026 – Dec 2026 Supporting students in the mathematical foundations of AI through office hours, coursework, and grading.
Software Engineering Intern — CodSoft Jan 2025 – Mar 2025 Shipped a Node/React app to AWS EC2 under pm2, containerized the backend with a multi-stage Dockerfile, migrated data to AWS RDS PostgreSQL, and automated deploys with a GitHub Actions workflow.
Software Engineering Intern — Cognifyz Technology Jul 2024 – Sep 2024 Built Express REST endpoints with JWT auth and Redis session storage, designed MongoDB compound indexes to speed up hot endpoints, and enforced Jest coverage and linting in a CI pipeline.

Figure 2. About page with experience and bio.

Projects page (AI-generated)

This is the most involved page. There is a sort dropdown and a row of technology filters, each showing how many projects use that tag. Every project is a card with an architecture diagram generated as SVG, a short description, its tech tags, an Architecture button that opens a detail dialog, and a link to the repo. Right now it shows four projects: SecureCheck, TransactIQ, PromptRelay, and HelmWarden.

Projects

Filter by technology and open any project to see how it is put together.

Featured first

AWS 1

Docker 4

Go 2

Java 1

Kafka 1

Kubernetes 2

Node.js 1

Prometheus 1

React 1

Redis 2

Security 2

Terraform 1

Showing 4 of 4 projects

PR

API GM

Lambda

DynamoDB

Semgrep

SQS

SecureCheck2026

CI/CD security gate that blocks risky merges

A CI/CD security gate that automatically scans every GitHub pull request for vulnerabilities and blocks merges on HIGH severity findings.

AWS

Terraform

Docker

Security

React

Node.js

Architecture

View on GitHub

API

Outbox

Kafka

Ledger

Fraud

Consumer

TransactIQ2025

Kafka payment pipeline with effectively-once processing

Kafka payment pipeline with a transactional outbox and idempotent consumers for effectively-once processing, plus an LLM-assisted fraud-triage service.

Java

Kafka

Kubernetes

Redis

Docker

Architecture

View on GitHub

Client

Gateway

Cache

Metrics

Provider

Router

PromptRelay2025

OpenAI-compatible gateway with a semantic cache

OpenAI-compatible Go gateway fronting multiple LLM providers with fallback, a semantic response cache over Redis, and Prometheus/Grafana observability.

Go

Redis

Docker

Prometheus

Architecture

View on GitHub

Figure 3. Projects gallery with filters, sort, and generated architecture diagrams.